

Other kinds of filters

Apart from these various special effect filters are used for added drama to pictures, available in a wide range of grades and useful for both colour and black & white images. Typically the optical effects all involve bending a percentage of image forming light from its original path to defocus it. The most commonly used filters in this category are **diffusion filters** and **cool and warm filters**. Diffusion filters soften subjects generating a dreamy blur, most often used in portraits. With such filters higher contrast scenes appear sharper, needing more diffusion than scenes with lower contrast. These requirements also vary with other conditions, like colour allows less diffusion than black & white, or like smaller file formats and large-screen projectors will allow lesser haze. Since the effect can be achieved in numerous ways, the diffusion filters from different manufacturers also vary significantly. The most common approaches are to use some kind of netting or mesh in the filter, which is also one of the earliest methods to accomplish the effect, and another way is to use something which is transparent but not optically sharp. When using gridding or netting technique, you can choose between various widths, colours, spacings and shapes (typically diamonds or squares). Nylon is popular choice for such meshes and the finer the mesh, more is the image area covered by mesh lines leading to stronger effects. The second most common way is **transparent diffusion**, where tiny globs of acrylic are deposited on one surface of the lens, allowing it to act as different micro-lenses to diffuse the light. Both these techniques can



Adjust colour tone on the fly

be digitally added to the images during post production, and these photo editing softwares in principle provide a very precise degree of control of the level of effect, however the overall look may be noticeably different.

Another popular option to play around with in photography are cooling and warming filters which change the white balance of light reaching the camera's sensors.